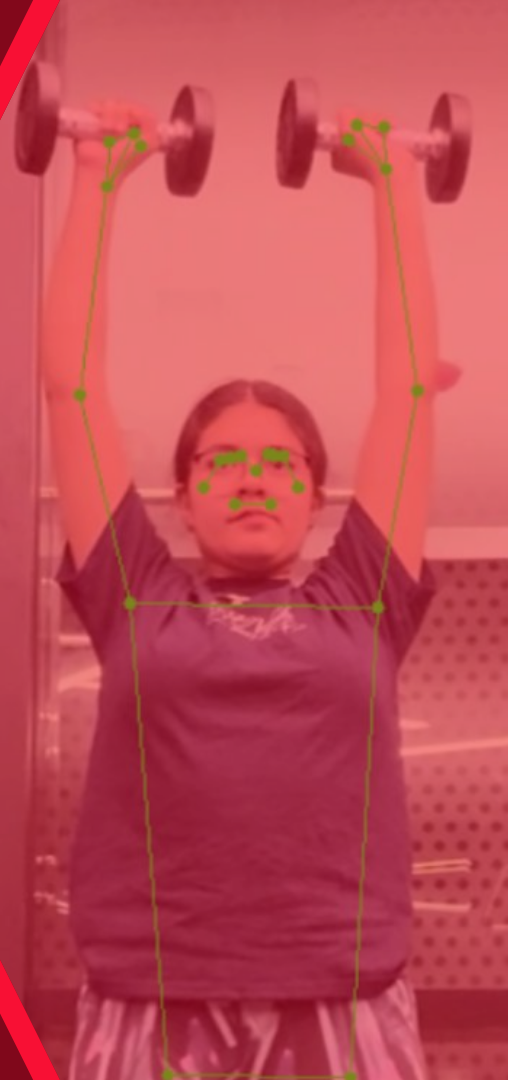


gAInz

CIS 5810, Final Project, FALL 2025

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THE PROBLEM



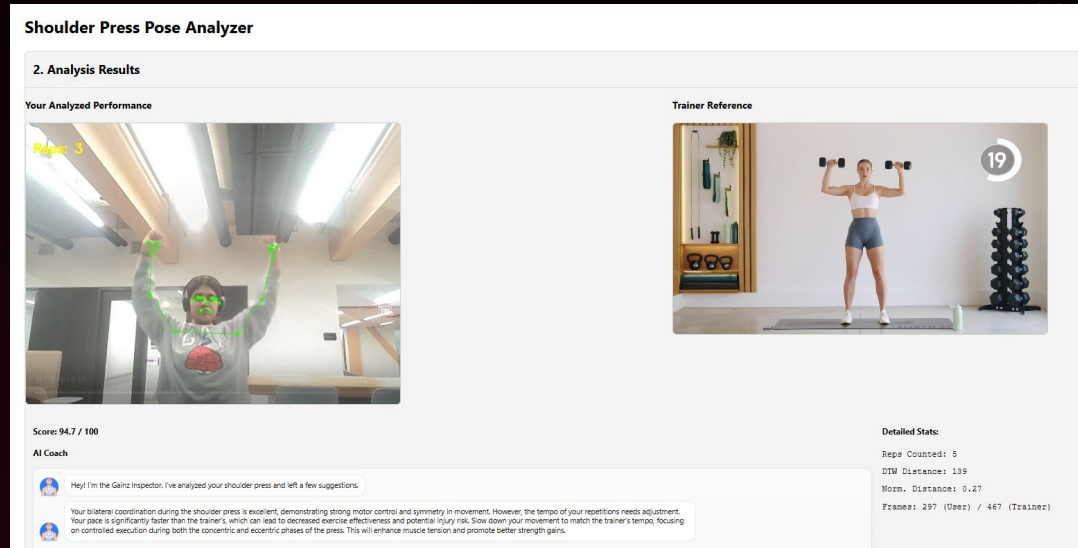
- When people work out and learn new exercises on their own, it is hard to know if their form matches what a trainer would recommend.
- Incorrect form → injury + slow progress
- Most fitness apps track your reps but don't evaluate the quality of your form



Can computer vision compare their motion to a trainer and give personalized feedback?

WHAT gAInz DOES

- Flask webapp where users record or upload an exercise video.
- Backend extracts skeletons for both user and trainer using BlazePose.
- Compares the two movements to identify form differences.
- Displays both videos side-by-side with skeleton overlays.
- Generates personalized feedback through an LLM-powered trainer character.

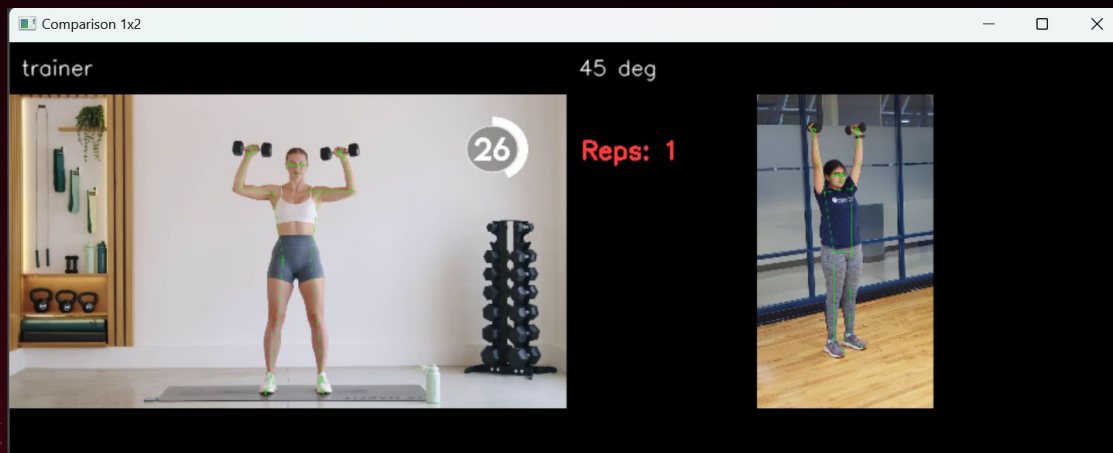


PHASE 1: CORE COMPARISON SYSTEM

- Chose the BlazePose model to track joints frame by frame, as the user moves.
- Extracted skeletons for both user and trainer.
- Measured simple joint angles (elbow, shoulder, hip, knee) during exercise.
- Compared depth and symmetry of the user movement to the trainer's.
- Produced our first basic form score.



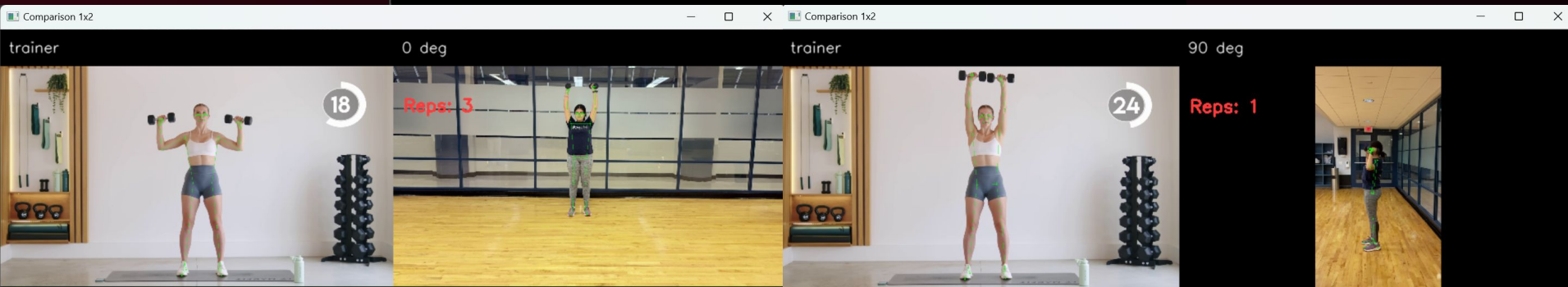
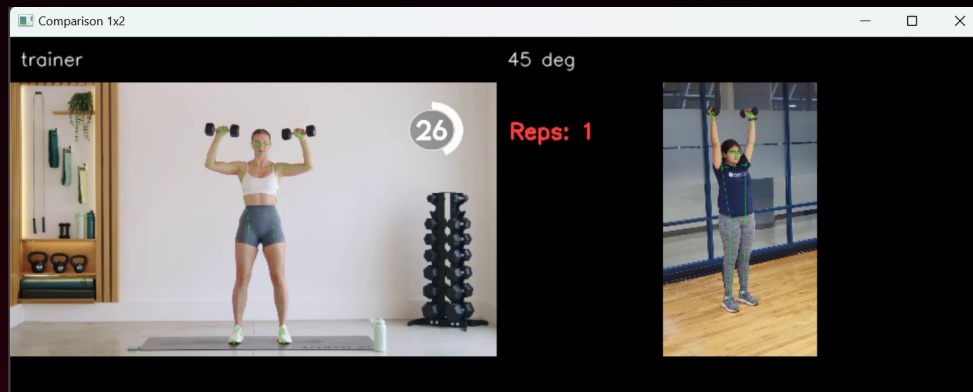
PHASE 2: MAKING THE SYSTEM ROBUST



- Users recorded from different angles and distances (0° , 45° , 90°) for prediction stability.
- We aligned user and trainer skeletons with a Kabsch transform to fix angle and scale differences.
- We used FastDTW to match pacing so reps could be compared even at different speeds or when videos were not perfectly matched.



Compared multiple perspectives



PHASE 3: UI DESIGN + AI FEEDBACK

- Integrated the pipeline into a Flask webapp.
- Collected the user's form differences after alignment at specific points in the video.
- Sent differences to an LLM for trainer-style feedback.
- Displayed videos, skeleton overlays, reps, and feedback in the UI.

Your Analyzed Performance


Trainer Reference


Score: 92 / 100
AI Coach


Hey! I'm the Gainz Inspector. I've analyzed your shoulder press and left a few suggestions.

Bilateral coordination during the shoulder press was excellent. However, improvements are needed in shoulder positioning; ensure both the left and right shoulders are stable and aligned with the torso throughout the movement. Additionally, the pace of the exercise was too fast; aim to match the tempo more closely with the trainer for optimal results.



Detailed Stats:
Reps Counted: 13
DTW Distance: 202.34
Norm. Distance: 0.4
Frames: 311 (User) / 467 (Trainer)

Record Again



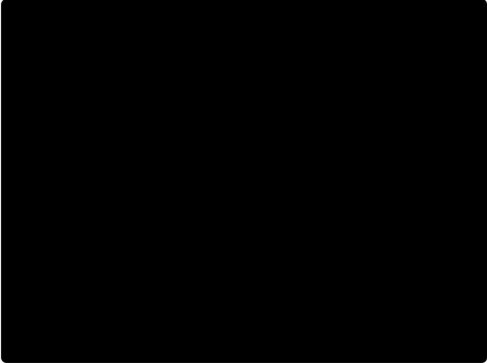
Phase 4: Expanding the Uses

gAlnz Form Inspector

[← Back](#)

Recording: Squat

Option 1: Record Live



[Start Recording](#)

Option 2: Upload Video

Have a pre-recorded video?

[Choose File](#) No file chosen

[Analyze Uploaded File](#)

Could not access camera. Please allow camera permissions.

- Create the comparison system for more exercises (squat, jumping jacks) using various key joints and positions
- Integrate the webapp to choose the desired exercise
- Allow for video upload or live recording

gAlnz Form Inspector

Choose Your Exercise

Shoulder Press

Barbell or Dumbbell overhead press.

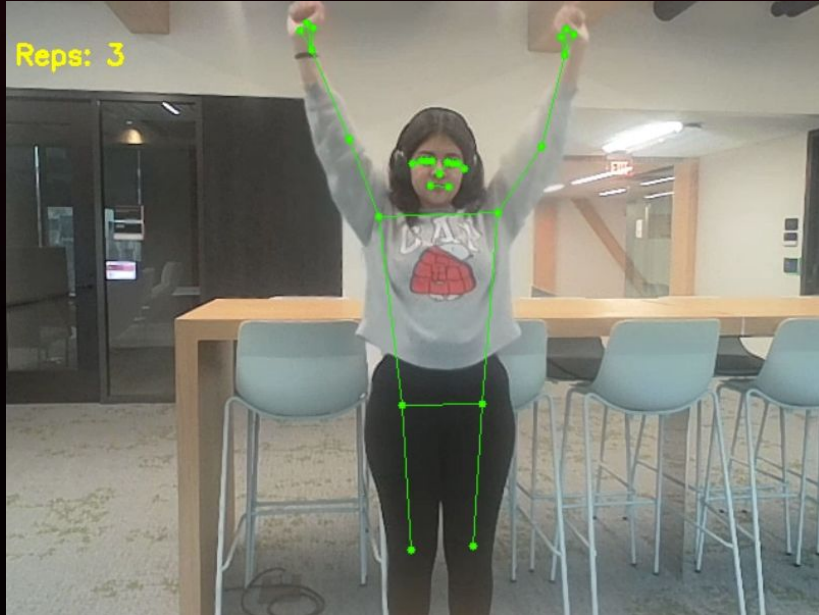
Squat

Standard bodyweight or weighted squat.

Jumping Jacks

Cardio coordination check.

RESULTS AND LIMITATIONS



What Our System Handles

- Works across different camera angles and distances than the trainer after alignment.
- Can compare reps when the user moves faster or slower than the trainer.
- It supports more than one exercise; we tested both upper and lower-body movements.

Limitations

- No real time feedback, processing must happen after recording
- LLM feedback is dependent on quality of the recording.





THANK YOU!

QUESTIONS?

